Indian Institute of Technology Bombay
Department of Electrical Engineering

Assignment-4
Physical Design of 16-bit Kogge-Stone Tree Adder

Course: EE671 - VLSI Design

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Contents

Chapter 1

Introduction

This report presents the **physical design implementation** of a 16-bit Kogge-Stone Tree Adder using **Cadence Innovus**. The RTL netlist was generated from Assignment-3, and the backend flow was completed using Innovus targeting the 45nm standard cell library.

1.1 Objective

- Perform ASIC physical design flow using Cadence Innovus.
- Generate layout and relevant design files: **GDSII, DEF, SPEF, SDF**.
- Verify: **DRC, Connectivity, Antenna violations** - all 0.
- Extract: **post-route timing, area, and power**.
- Perform **post-layout simulation** using SDF back-annotation.

Chapter 2

Physical Design Flow

The physical design steps followed in Innovus are:

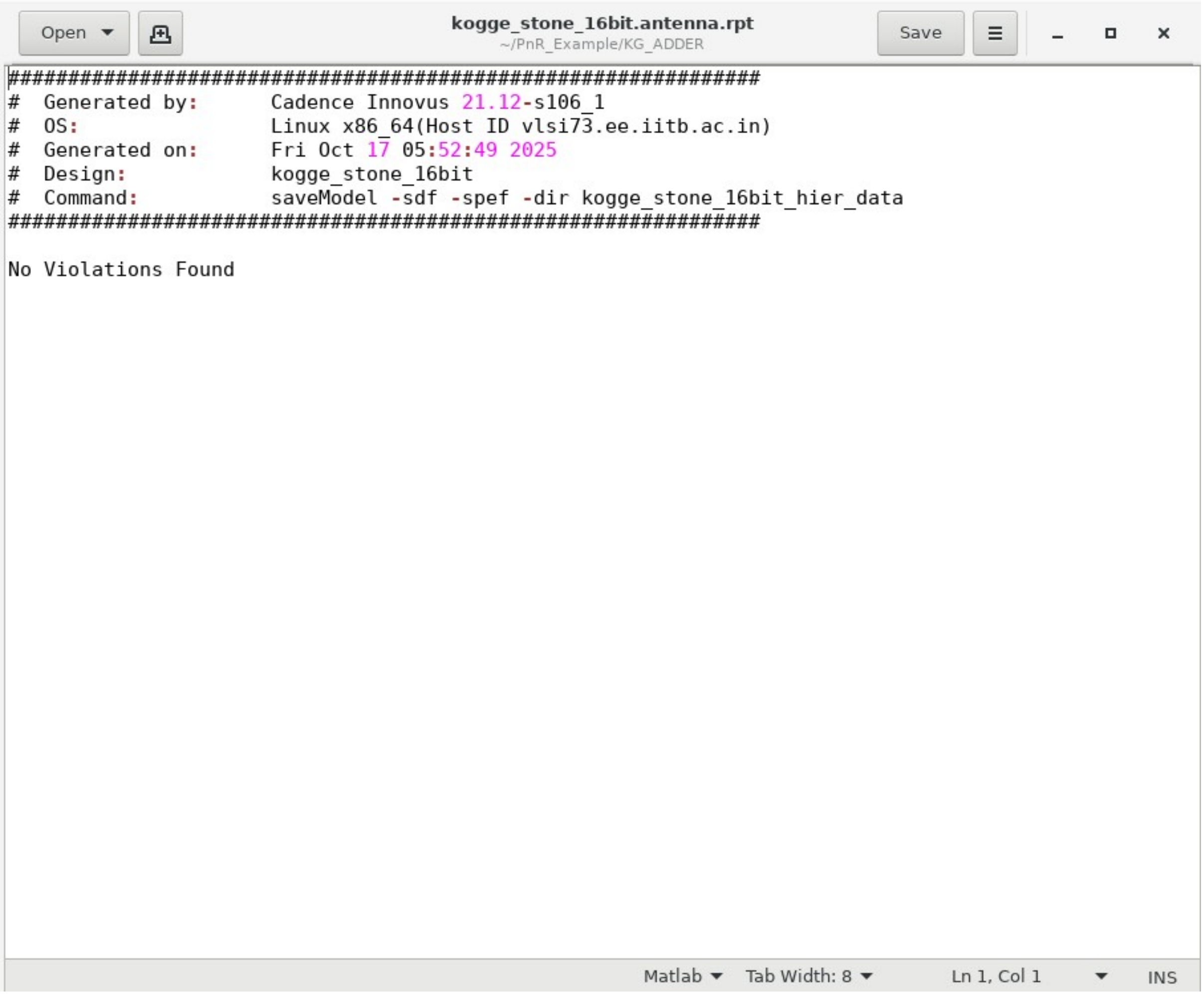
1. Floorplanning
2. Power Planning
3. Placement
4. Clock Tree Synthesis (CTS)
5. Routing
6. RC Extraction and Optimization
7. Timing Closure
8. Physical Verification and Signoff

Chapter 3

DRC, Antenna, and Connectivity Reports

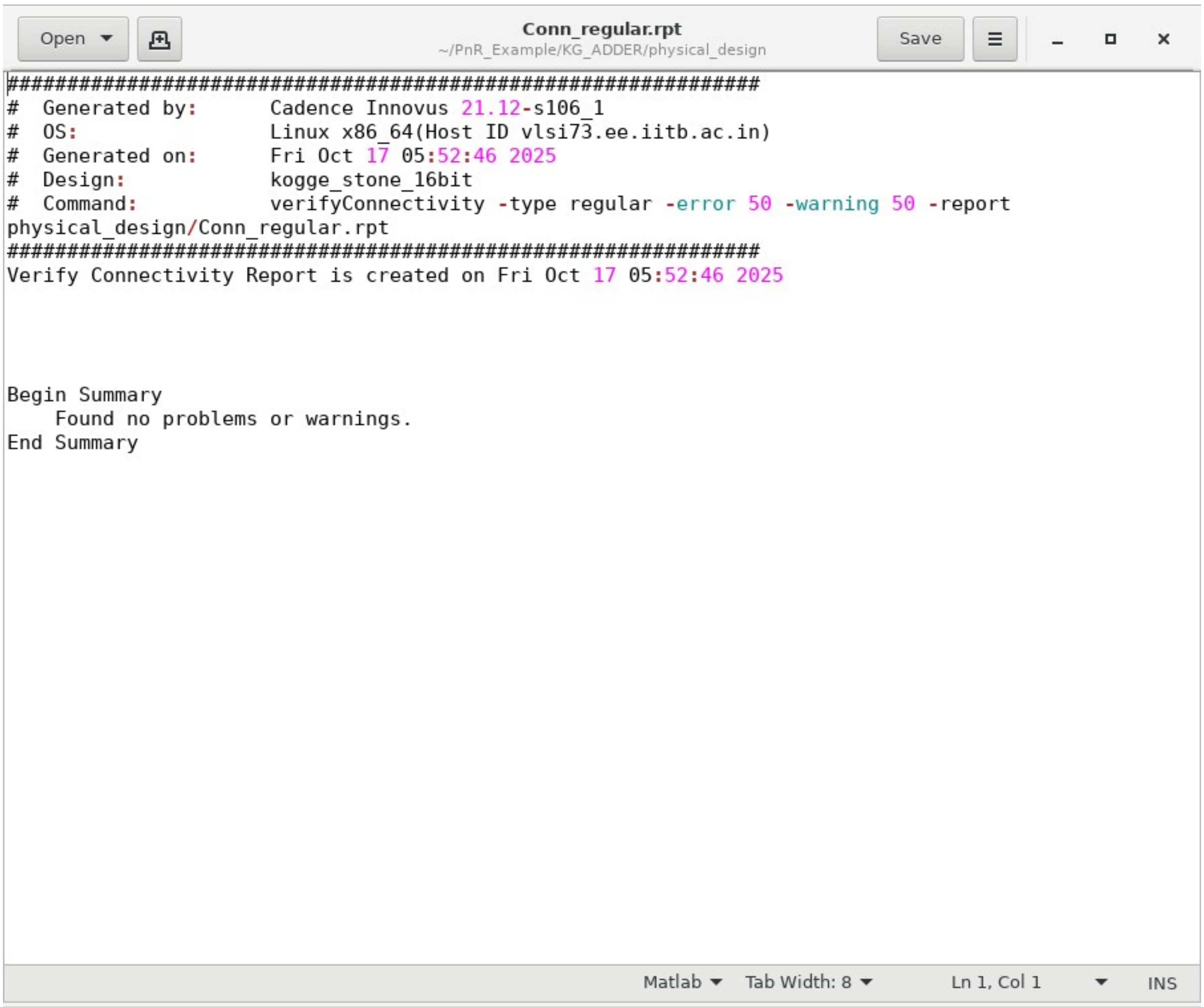
```
Output file is physical_design/power.rpt
#-check_same_via_cell true               # bool, default=false, user setting
 *** Starting Verify DRC (MEM: 1743.6) ***

  VERIFY DRC ...... Starting Verification
  VERIFY DRC ...... Initializing
  VERIFY DRC ...... Deleting Existing Violations
  VERIFY DRC ...... Creating Sub-Areas
  VERIFY DRC ...... Using new threading
  VERIFY DRC ...... Sub-Area: {0.000 0.000 57.600 57.200} 1 of 1
  VERIFY DRC ...... Sub-Area : 1 complete 0 Viols.

  Verification Complete : 0 Viols.

 *** End Verify DRC (CPU: 0:00:00.0  ELAPSED TIME: 0.00  MEM: 4.0M) ***

VERIFY_CONNECTIVITY use new engine.

******** Start: VERIFY CONNECTIVITY ********
Start Time: Fri Oct 17 05:52:46 2025

Design Name: kogge_stone_16bit
Database Units: 2000
Design Boundary: (0.0000, 0.0000) (57.6000, 57.2000)
Error Limit = 50; Warning Limit = 50
Check specified nets

Begin Summary
  Found no problems or warnings.
End Summary

End Time: Fri Oct 17 05:52:46 2025
Time Elapsed: 0:00:00.0

******** End: VERIFY CONNECTIVITY ********
  Verification Complete : 0 Viols.  0 Wrngs.
  (CPU Time: 0:00:00.0  MEM: 0.000M)
```

Figure 3.1: DRC Report - 0 violations




```
################################################################
#  Generated by:      Cadence Innovus 21.12-s106_1
#  OS:                Linux x86_64(Host ID vlsi73.ee.iitb.ac.in)
#  Generated on:      Fri Oct 17 05:52:49 2025
#  Design:            kogge_stone_16bit
#  Command:           saveModel -sdf -spef -dir kogge_stone_16bit_hier_data
################################################################

No Violations Found
```

Figure 3.2: Antenna Report - 0 violations



```
################################################################
#  Generated by:      Cadence Innovus 21.12-s106_1
#  OS:                Linux x86_64(Host ID vlsi73.ee.iitb.ac.in)
#  Generated on:      Fri Oct 17 05:52:46 2025
#  Design:            kogge_stone_16bit
#  Command:           verifyConnectivity -type regular -error 50 -warning 50 -report
physical_design/Conn_regular.rpt
################################################################
Verify Connectivity Report is created on Fri Oct 17 05:52:46 2025



Begin Summary
    Found no problems or warnings.
End Summary
```

Figure 3.3: Connectivity Report - All nets connected correctly

Chapter 4

Final Layout

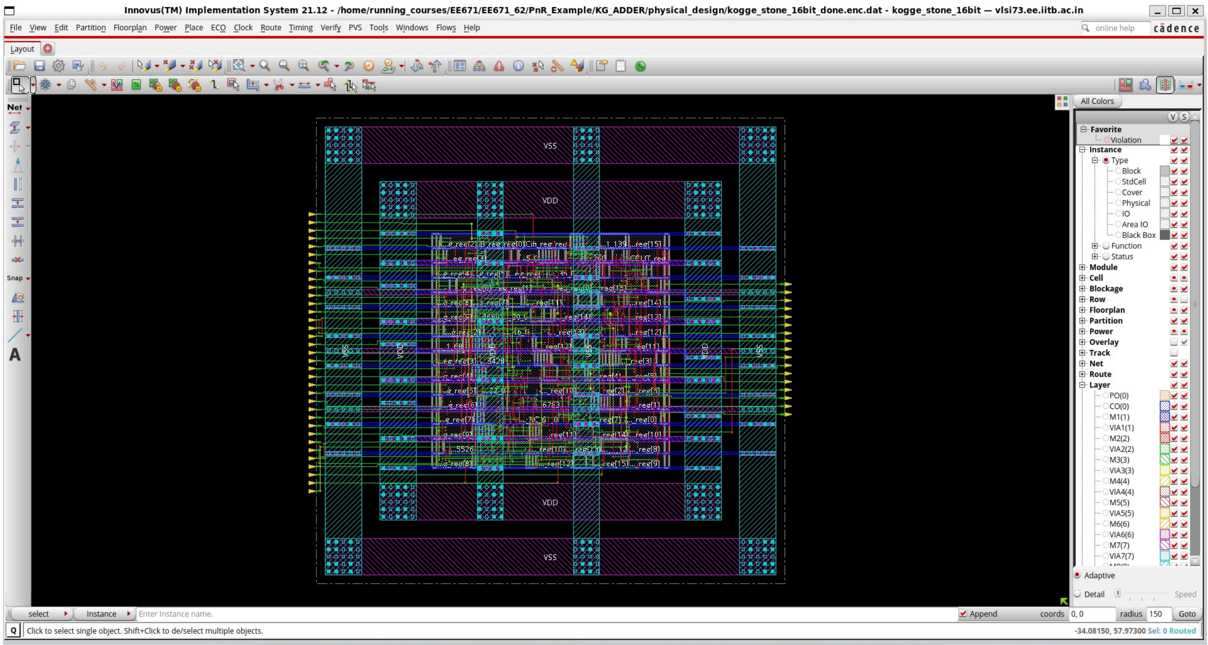


Figure 4.1: Final Routed Layout of 16-bit Kogge-Stone Adder

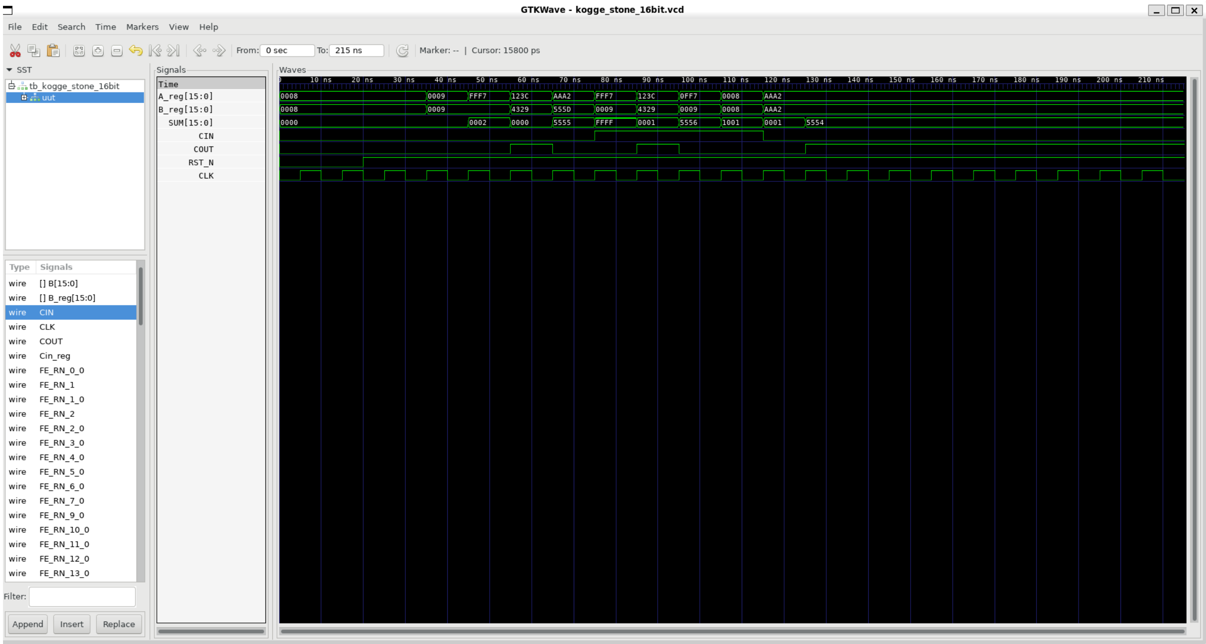


Figure 4.2: Post Layout Kogge-Stone Adder Output Waveform(GTKwave)

Chapter 5

Post-Route Timing, Area, and Power

```
Setup views included:
 WORST_CASE

+--------------------+---------+---------+---------+
|     Setup mode     |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.003  |  0.003  |  0.871  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   100   |   17    |   83    |
+--------------------+---------+---------+---------+

+----------------+-------------------------------+------------------+
|                |              Real             |       Total      |
|    DRVs        +------------------+------------+------------------|
|                |  Nr nets(terms)  | Worst Vio  |  Nr nets(terms)  |
+----------------+------------------+------------+------------------+
|   max_cap      |      0 (0)       |   0.000    |      0 (0)       |
|   max_tran     |      0 (0)       |   0.000    |      0 (0)       |
|   max_fanout   |      0 (0)       |     0      |      0 (0)       |
|   max_length   |      0 (0)       |     0      |      0 (0)       |
+----------------+------------------+------------+------------------+

Density: 100.000%
------------------------------------------------------------------
Reported timing to dir ./timingReports
Total CPU time: 1.14 sec
Total Real time: 3.0 sec
Total Memory Usage: 1745.148438 Mbytes
*** timeDesign #4 [finish] : cpu/real = 0:00:01.1/0:00:03.3 (0.3), totSession cpu/real = 0:01:23.7/0:01:40.5 (0.8), mem = 1745.1M
###############################################################
#  Generated by:      Cadence Innovus 21.12-s106_1
#  OS:                Linux x86_64(Host ID vlsi73.ee.iitb.ac.in)
#  Generated on:      Fri Oct 17 05:52:45 2025
#  Design:            kogge_stone_16bit
#  Command:           report_timing
###############################################################
Path 1: MET Setup Check with Pin SUM_reg[15]/CP
Endpoint:   SUM_reg[15]/D   (^) checked with  leading edge of 'CLK'
Beginpoint: B_reg_reg[1]/QN (^) triggered by  leading edge of 'CLK'
Path Groups: {CLK}
Analysis View: WORST_CASE
Other End Arrival Time          0.002
- Setup                         0.032
+ Phase Shift                   1.000
= Required Time                 0.970
- Arrival Time                  0.967
= Slack Time                    0.003
     Clock Rise Edge                      0.000
     + Clock Network Latency (Prop)  0.000
     = Beginpoint Arrival Time       0.000
```

Figure 5.1: Setup slack report - snapshot 1

```
Path 1: MET Setup Check with Pin SUM_reg[15]/CP
Endpoint:   SUM_reg[15]/D   (^) checked with  leading edge of 'CLK'
Beginpoint: B_reg_reg[1]/QN (^) triggered by  leading edge of 'CLK'
Path Groups: {CLK}
Analysis View: WORST_CASE
Other End Arrival Time          0.002
- Setup                         0.032
+ Phase Shift                   1.000
= Required Time                 0.970
- Arrival Time                  0.967
= Slack Time                    0.003
     Clock Rise Edge                 0.000
     + Clock Network Latency (Prop)  0.000
     = Beginpoint Arrival Time       0.000
     +---------------------------------------------------------------------------+
     |   Instance    |     Arc      |   Cell    | Delay | Arrival | Required |
     |               |              |           |       |  Time   |   Time   |
     |---------------+--------------+-----------+-------+---------+----------|
     | B_reg_reg[1]  | CP ^         |           |       |   0.000 |    0.003 |
     | B_reg_reg[1]  | CP ^ -> QN ^ | DFCND1    | 0.185 |   0.185 |    0.188 |
     | FE_RC_45_0    | B1 ^ -> ZN v | OAI22D2   | 0.042 |   0.228 |    0.230 |
     | FE_RC_38_0    | A2 v -> ZN ^ | CKND2D1   | 0.044 |   0.271 |    0.274 |
     | FE_RC_37_0    | B ^ -> ZN v  | IOA21D2   | 0.042 |   0.314 |    0.317 |
     | FE_RC_20_0    | A1 v -> ZN ^ | AOI21D4   | 0.049 |   0.363 |    0.366 |
     | FE_RC_46_0    | A1 ^ -> ZN v | OAI21D4   | 0.032 |   0.395 |    0.398 |
     | FE_RC_14_0    | A1 v -> ZN ^ | ND2D2     | 0.028 |   0.424 |    0.426 |
     | FE_RC_13_0    | A1 ^ -> ZN v | ND2D2     | 0.029 |   0.452 |    0.455 |
     | FE_RC_31_0    | A1 v -> ZN ^ | CKND2D2   | 0.027 |   0.479 |    0.482 |
     | FE_RC_30_0    | B ^ -> ZN v  | IOA21D2   | 0.037 |   0.516 |    0.519 |
     | FE_RC_22_0    | B1 v -> ZN ^ | IND2D4    | 0.029 |   0.545 |    0.548 |
     | FE_RC_61_0    | A2 ^ -> ZN v | OAI211D4  | 0.042 |   0.587 |    0.590 |
     | FE_RC_56_0    | B1 v -> ZN ^ | IND2D2    | 0.034 |   0.620 |    0.623 |
     | FE_RC_55_0    | A1 ^ -> ZN v | ND2D2     | 0.027 |   0.648 |    0.651 |
     | FE_RC_1_0     | B1 v -> ZN ^ | IND2D1    | 0.027 |   0.675 |    0.678 |
     | FE_RC_0_0     | A1 ^ -> ZN v | CKND2D2   | 0.029 |   0.704 |    0.707 |
     | FE_RC_26_0    | A1 v -> ZN ^ | CKND2D2   | 0.026 |   0.730 |    0.732 |
     | FE_RC_25_0    | A1 ^ -> ZN v | ND2D2     | 0.032 |   0.762 |    0.764 |
     | FE_RC_29_0    | A1 v -> ZN ^ | CKND2D2   | 0.028 |   0.789 |    0.792 |
     | FE_RC_28_0    | B ^ -> ZN v  | IOA21D2   | 0.031 |   0.820 |    0.823 |
     | FE_RC_33_0    | A1 v -> ZN ^ | CKND2D2   | 0.027 |   0.847 |    0.850 |
     | FE_RC_32_0    | B ^ -> ZN v  | IOA21D2   | 0.030 |   0.878 |    0.881 |
     | FE_RC_35_0    | A1 v -> ZN ^ | ND2D2     | 0.027 |   0.905 |    0.908 |
     | FE_RC_50_0    | A2 ^ -> ZN ^ | IOA21D2   | 0.062 |   0.967 |    0.970 |
     | SUM_reg[15]   | D ^          | DFCNQD1   | 0.000 |   0.967 |    0.970 |
     +---------------------------------------------------------------------------+

*** timeDesign #5 [begin] : totSession cpu/real = 0:01:23.8/0:01:40.5 (0.8), mem = 1729.4M
```

Figure 5.2: Setup slack report - snapshot 2

```
        timeDesign Summary
-------------------------------------------------------------------

Hold views included:
 BEST_CASE

+--------------------+---------+---------+---------+
|     Hold mode      |   all   | reg2reg | default |
+--------------------+---------+---------+---------+
|           WNS (ns):|  0.050  |  0.073  |  0.050  |
|           TNS (ns):|  0.000  |  0.000  |  0.000  |
|    Violating Paths:|    0    |    0    |    0    |
|          All Paths:|   100   |   17    |   83    |
+--------------------+---------+---------+---------+

Density: 100.000%
-------------------------------------------------------------------
Reported timing to dir ./timingReports
Total CPU time: 0.3 sec
Total Real time: 1.0 sec
Total Memory Usage: 1731.578125 Mbytes
*** timeDesign #5 [finish] : cpu/real = 0:00:00.3/0:00:00.3 (1.0), totSession cpu/real = 0:01:24.1/0:01:40.8 (0.8), mem = 1731.6M
###############################################################
#  Generated by:      Cadence Innovus 21.12-s106_1
#  OS:                Linux x86_64(Host ID vlsi73.ee.iitb.ac.in)
#  Generated on:      Fri Oct 17 05:52:46 2025
#  Design:            kogge_stone_16bit
#  Command:           report_timing
###############################################################
Path 1: MET Removal Check with Pin A_reg_reg[8]/CP
Endpoint:   A_reg_reg[8]/CDN (^) checked with  leading edge of 'CLK'
Beginpoint: RST_N            (^) triggered by  leading edge of 'CLK'
Path Groups: {async_default}
Analysis View: BEST_CASE
Other End Arrival Time          0.002
+ Removal                       0.049
+ Phase Shift                   0.000
= Required Time                 0.051
  Arrival Time                  0.101
  Slack Time                    0.050
     Clock Rise Edge                      0.000
     + Input Delay                        0.100
     = Beginpoint Arrival Time            0.100
     +-----------------------------------------------------------------+
     |   Instance   |   Arc   |  Cell   | Delay | Arrival | Required |
     |              |         |         |       |  Time   |   Time   |
     |--------------+---------+---------+-------+---------+----------|
     |              | RST_N ^ |         |       |   0.100 |    0.050 |
     | A_reg_reg[8] | CDN ^   | DFCNQD1 | 0.001 |   0.101 |    0.051 |
     +-----------------------------------------------------------------+
```

Figure 5.3: Hold slack report

```
area.rpt
~/PnR_Example/KG_ADDER/physical_design

     Hinst Name          Module Name          Inst Count          Total Area
--------------------------------------------------------------------------
kogge_stone_16bit                                   141             665.280
```

Figure 5.4: Area report

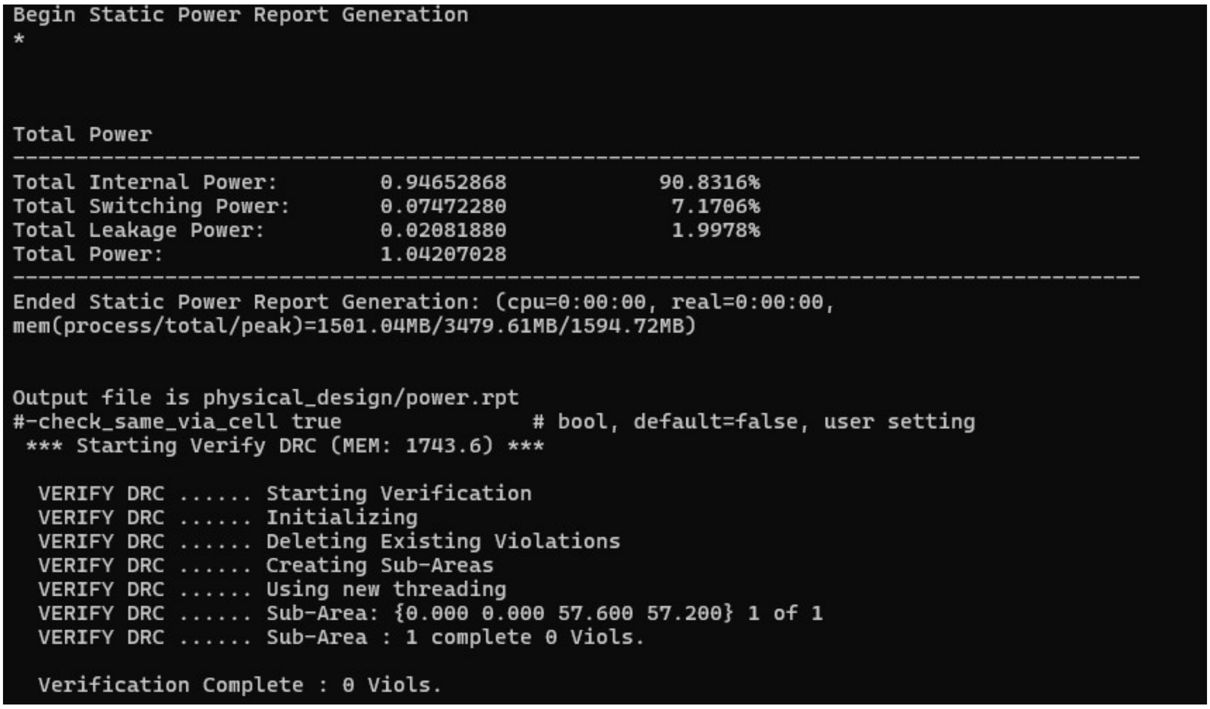
Open power.rpt
~/PnR_Example/KG_ADDER/physical_design

```
*----------------------------------------------------------------------------------
*       Innovus 21.12-s106_1 (64bit) 12/08/2021 18:19 (Linux 3.10.0-693.el7.x86_64)
*
*
*       Date & Time:    2025-Oct-17 05:52:46 (2025-Oct-17 00:22:46 GMT)
*
*----------------------------------------------------------------------------------
*
*       Design: kogge_stone_16bit
*
*       Liberty Libraries used:
*               BEST_CASE: /vlsi/pdk/course_pdk_2025/tsmc_gp_65_stdio/tcbn65gplus_200a/TSMCHOME/digital/Front_End/timing_power_noise/CCS/tcbn65gplus_200a/tcbn65gplusbc_ccs.lib
*
*       Power Domain used:
*               Rail:        VDD      Voltage:        1.1
*
*       Power View : BEST_CASE
*
*       User-Defined Activity : N.A.
*
*       Activity File: N.A.
*
*       Hierarchical Global Activity: N.A.
*
*       Global Activity: N.A.
*
*       Sequential Element Activity: 0.200000
*
*       Primary Input Activity: 0.200000
*
*       Default icg ratio: N.A.
*
*       Global Comb ClockGate Ratio: N.A.
*
*       Power Units = 1mW
*
*       Time Units = 1e-09 secs
*
*       report_power -outfile physical_design/power.rpt
*
-----------------------------------------------------------------------------------


Total Power
-----------------------------------------------------------------------------------
Total Internal Power:       0.94652868           90.8316%
Total Switching Power:      0.07472280            7.1706%
Total Leakage Power:        0.02081880            1.9978%
Total Power:                1.04207028
-----------------------------------------------------------------------------------
```

Figure 5.5: Power Consumption report snapshot 1

```
-----------------------------------------------------------------------------------------


Group                           Internal   Switching    Leakage      Total    Percentage
                                Power      Power        Power        Power    (%)
-----------------------------------------------------------------------------------------
Sequential                        0.8592     0.02201   0.009217     0.8905       85.45
Macro                                  0           0          0          0           0
IO                                     0           0          0          0           0
Physical-Only                          0           0   0.003517   0.003517      0.3375
Combinational                    0.08728     0.05271   0.008085     0.1481       14.21
Clock (Combinational)                  0           0          0          0           0
Clock (Sequential)                     0           0          0          0           0
-----------------------------------------------------------------------------------------
Total                             0.9465     0.07472    0.02082      1.042         100
-----------------------------------------------------------------------------------------


Rail                  Voltage   Internal   Switching    Leakage      Total    Percentage
                                Power      Power        Power        Power    (%)
-----------------------------------------------------------------------------------------
VDD                       1.1     0.9465     0.07472    0.02082      1.042         100


-----------------------------------------------------------------------------------------
*       Power Distribution Summary:
*               Highest Average Power:              A_reg_reg[0] (DFCND1):          0.01906
*               Highest Leakage Power:              B_reg_reg[0] (DFCNQD4):       0.0003167
*           Total Cap:      4.86544e-13 F
*           Total instances in design:   201
*           Total instances in design with no power:     0
*           Total instances in design with no activity:     0
*           Total Fillers and Decap:    60
-----------------------------------------------------------------------------------------
```

Figure 5.6: Power Consumption report snapshot 2



```
Begin Static Power Report Generation
*



Total Power
-----------------------------------------------------------------------------------------
Total Internal Power:        0.94652868           90.8316%
Total Switching Power:       0.07472280            7.1706%
Total Leakage Power:         0.02081880            1.9978%
Total Power:                 1.04207028
-----------------------------------------------------------------------------------------
Ended Static Power Report Generation: (cpu=0:00:00, real=0:00:00,
mem(process/total/peak)=1501.04MB/3479.61MB/1594.72MB)


Output file is physical_design/power.rpt
#-check_same_via_cell true              # bool, default=false, user setting
 *** Starting Verify DRC (MEM: 1743.6) ***

  VERIFY DRC ...... Starting Verification
  VERIFY DRC ...... Initializing
  VERIFY DRC ...... Deleting Existing Violations
  VERIFY DRC ...... Creating Sub-Areas
  VERIFY DRC ...... Using new threading
  VERIFY DRC ...... Sub-Area: {0.000 0.000 57.600 57.200} 1 of 1
  VERIFY DRC ...... Sub-Area : 1 complete 0 Viols.

  Verification Complete : 0 Viols.
```

Figure 5.7: Total Power Consumption report

Table 5.1: Post-route timing, area, and power results

Parameter	**Value**
Clock frequency used for synthesis (MHz)	100
Worst case setup slack [WNS in the report] (ns)	0.003
Worst case hold slack [WHS in the report] (ns)	0.050
Design area (μm^2)	665.28
Power Consumption (Sequential Only) (μW)	0.8905
Power Consumption (Combinational Only) (μW)	0.1481
Total Power Consumption (Internal Only) (μW)	0.94652868
Total Power Consumption (Switching Only) (μW)	0.07472280
Total Power Consumption (Leakage Only) (μW)	0.02081880
Total Power Consumption (μW)	1.04207028

Chapter 6

Post-Layout Simulation

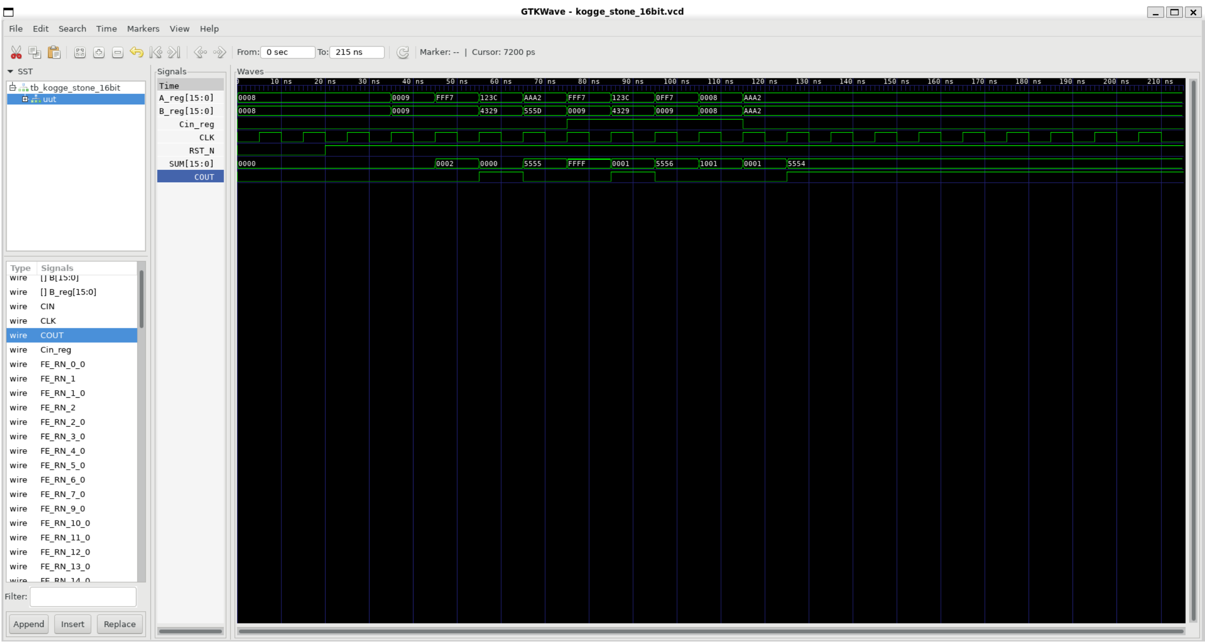


Figure 6.1: Post-Layout Functional Waveform (GTKWave)

Chapter 7

Conclusion

Physical design of the 16-bit Kogge-Stone Adder was successfully completed using Cadence Innovus. The final layout meets timing, area and power constraints, and has 0 DRC/Antenna violations. SDF-based simulation verified functional correctness.